

VARUN KUMAR R

Junior Data Science & AI Engineer

📍 Liverpool, UK • Open to Relocation: UK/EU/India

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PROFESSIONAL SUMMARY

A motivated fresher and MSc Data Science & Artificial Intelligence candidate at the University of Liverpool (Expected 2026), with a strong academic foundation in machine learning, natural language processing, deep learning, and data visualisation. Holds a B.Tech in Electronics & Communication Engineering with a Data Science specialisation (CGPA: 8.0) from SRM Institute of Science and Technology. Delivered five end-to-end AI and data science projects spanning computer vision, NLP, predictive modelling, IoT, and customer analytics, with original research presented at a National Conference (2024). Proficient in Python, TensorFlow, PyTorch, Scikit-learn, SQL, and Power BI. Seeking a Junior Data Science & AI Engineering role to apply rigorous analytical and engineering skills to real-world, impactful problems.

TECHNICAL SKILLS

Programming Languages Python, SQL, C, C++

ML / AI Frameworks TensorFlow, PyTorch, Scikit-learn

NLP & Computer Vision CNN Architectures, Sentiment Analysis, Text Classification

Data & Visualisation Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau

Soft Skills Analytical Thinking, Problem-solving, Communication, Collaboration, Adaptability

INTERNSHIP EXPERIENCE

Python & Machine Learning Engineering Intern

Kaashiv InfoTech, Chennai, India

Jun 2022 – Aug 2022

Technologies: Python, Scikit-learn, Pandas, NumPy, Automation Scripting

- Engineered Python automation scripts to eliminate repetitive data-entry workflows, improving overall team operational efficiency by **30%** and freeing analyst capacity for higher-value activities.
- Designed, trained, and validated supervised machine learning prototypes on structured business datasets, surfacing predictive insights that directly supported senior-level decision-making.
- Collaborated with senior engineers to debug, refactor, and optimise production-grade code, strengthening software quality and reducing average issue-resolution cycles.

EDUCATION

MSc — Data Science & Artificial Intelligence

University of Liverpool, United Kingdom

Expected 2026

B.Tech — Electronics & Communication Engineering

SRM Institute of Science & Technology, Chennai, India

2018 – 2022

Specialisation: Data Science CGPA: 8.0/10

ACADEMIC PROJECTS

Telecom Customer Churn Prediction

Technologies: Python, Random Forest, Scikit-learn, Pandas, Matplotlib

- Developed a multi-model classification pipeline using Random Forest on telecom subscriber data, achieving

86% accuracy and enabling data-driven proactive customer retention strategies.

- Performed comprehensive feature engineering and hyperparameter tuning, reducing model overfitting and improving generalisation across validation sets.

Plant Disease Detection — Computer Vision

Technologies: TensorFlow, PyTorch, Convolutional Neural Networks, OpenCV, Python

- Designed and trained a CNN image-classification model achieving **approximately 90% accuracy** on field crop imagery, enabling early disease diagnosis and reducing projected crop loss.
- Implemented data augmentation techniques including rotation, flipping, and zoom to improve model robustness on limited agricultural datasets.

Twitter Sentiment Analysis — NLP Pipeline

Technologies: Python, Scikit-learn, TF-IDF, Supervised Learning

- Built an end-to-end NLP pipeline classifying over **50,000 tweets**, achieving an **82% F1-score** and delivering actionable real-time brand sentiment monitoring insights.
- Applied tokenisation, stopword removal, and TF-IDF vectorisation to transform raw social media text into structured model-ready features.

E-commerce Customer Segmentation & Prediction

Technologies: Python, K-Means, Hierarchical Clustering, Scikit-learn, Seaborn, Power BI

- Applied K-Means and hierarchical clustering to segment e-commerce customers with **approximately 85% cluster accuracy**, improving targeted marketing effectiveness and retention by **15–20%**.
- Visualised customer segment profiles using Power BI dashboards, enabling non-technical stakeholders to interpret and act on findings.

IoT-Based Air Quality Monitoring System

Technologies: Python, IoT Sensors, ML Regression, Real-time Data Pipelines

- Engineered a sensor-integrated ML system to ingest and predict real-time pollution levels from IoT data streams, achieving strong predictive performance on live environmental data.
- Presented research findings at a **National Conference (2024)**, demonstrating applied AI for environmental health monitoring and public safety.

CERTIFICATIONS

- Data Science & AI Certification (6 months) — Boston Institute of Analytics
- Python for Data Science & Machine Learning
- Deep Learning in Neural Networks & AI with ChatGPT
- Power BI, RPA & Big Data Workshops (available upon request)

ACHIEVEMENTS & LEADERSHIP

- Presented research paper ‘IoT-based Air Quality Monitoring System’ at a National Conference (2024)
- Symposium Coordinator, Abhigyan ’22 — SRM University (2022)
- Joint Treasurer, Rotaract Club of SRM Vadapalani (2021 – 2023)

LANGUAGES

English Full Professional Proficiency **Tamil** Native Proficiency

DECLARATION

I hereby declare that all information furnished in this curriculum vitae is true and correct to the best of my knowledge and belief. I take full responsibility for the accuracy of the details provided above.

Varun Kumar R
Liverpool, United Kingdom